

The higher institute of engineering department of communication engineering

4th communication (Computer)

Network

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Experiment (7) Design and Implementation of an Integrated Network5

1- Objectives

learn how to make IP subnetting distribution in Shroud Academy

Step 1. Find host range

Step 2. Convert to binary

Step 3. Calculate the total number of hosts per subnet

Step 4. Calculate the number or subnets

Step 5. Verify the total number or subnets

2- Equipment / Program Requirements:

- a. End devices
- b. Switches
- c. router
- d. Straight or crossover cables
- e. Cisco Packet tracer

3- Simulation Results:

1- Router Configuration

```
Router0

Physical Config CLI Attributes

IOS Command Line Interface

%LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet4/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet4/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet5/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet5/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet6/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet6/0, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet7/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet7/0, changed state to up

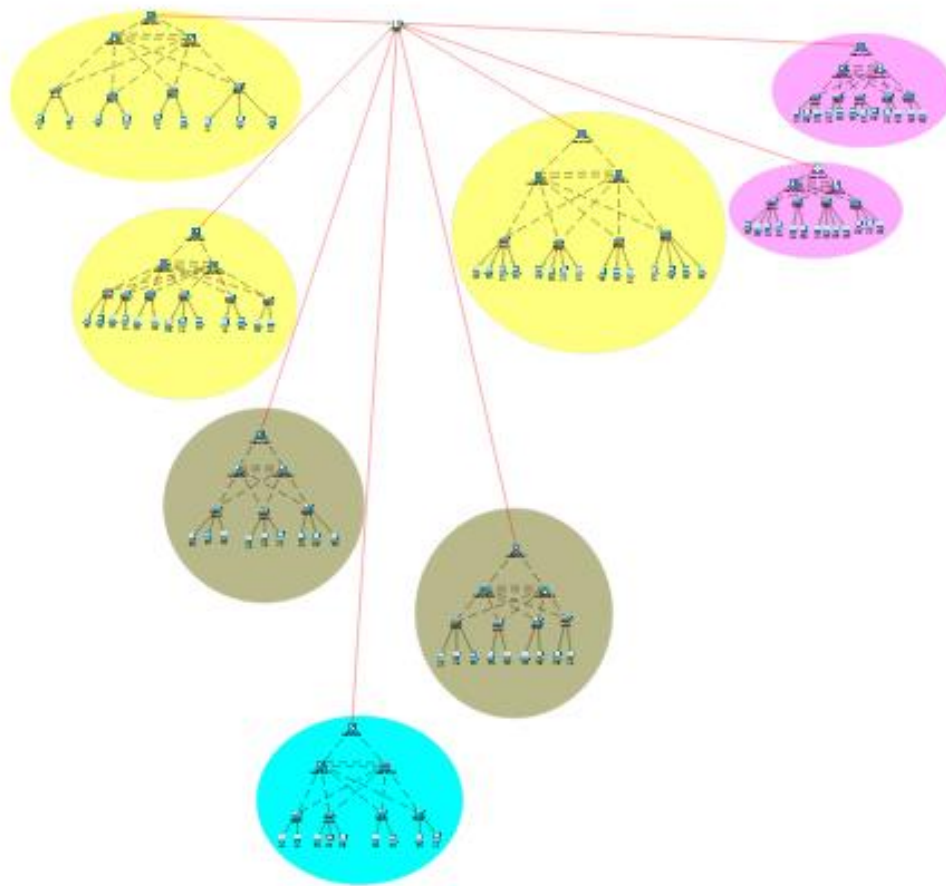
Router>enable
Router#Show ip interface brief
^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       192.168.3.129   YES manual up                    up
GigabitEthernet1/0       192.168.3.1     YES manual up                    up
GigabitEthernet2/0       192.168.2.129   YES manual up                    up
GigabitEthernet3/0       192.168.2.1     YES manual up                    up
GigabitEthernet4/0       192.168.1.129   YES manual up                    up
GigabitEthernet5/0       192.168.1.1     YES manual up                    up
GigabitEthernet6/0       192.168.0.129   YES manual up                    up
GigabitEthernet7/0       192.168.0.1     YES manual up                    up
GigabitEthernet8/0       unassigned      YES unset  administratively down  down
GigabitEthernet9/0       unassigned      YES unset  administratively down  down
Router#

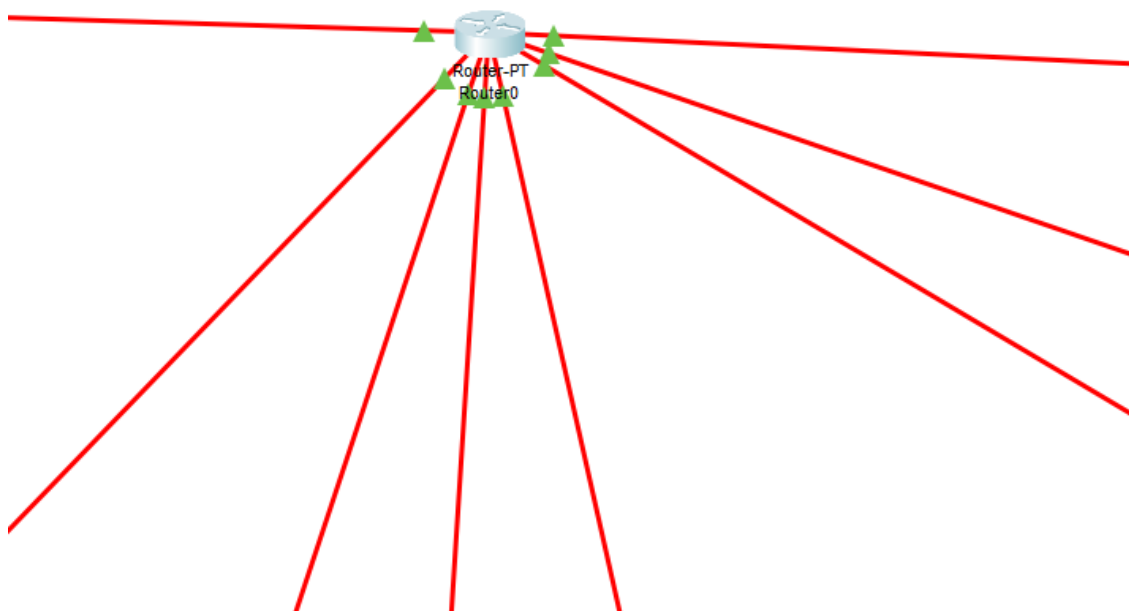
Copy Paste

Top
```

2- The entire network



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC77	PC79	ICMP		0.000	N	10	(ec ^
	Successful	PC78	PC86	ICMP		0.000	N	11	(ec
	Successful	PC31	PC41	ICMP		0.000	N	12	(ec v



PC54

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.1.137

Pinging 192.168.1.137 with 32 bytes of data:

Reply from 192.168.1.137: bytes=32 time<1ms TTL=128

Reply from 192.168.1.137: bytes=32 time<1ms TTL=128

Reply from 192.168.1.137: bytes=32 time<1ms TTL=128

Reply from 192.168.1.137: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.137:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

Command Prompt

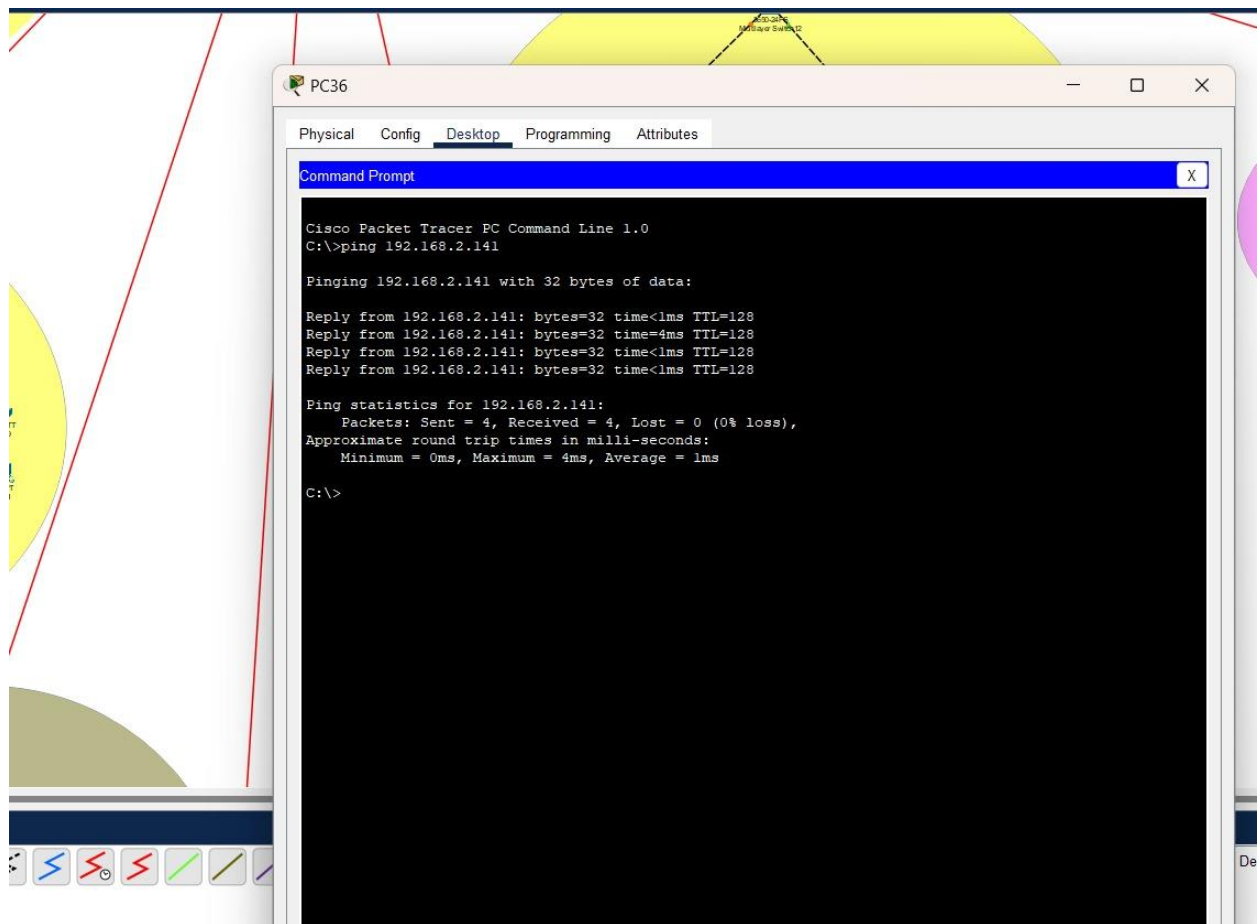
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.131

Pinging 192.168.3.131 with 32 bytes of data:

Reply from 192.168.3.131: bytes=32 time<1ms TTL=128
Reply from 192.168.3.131: bytes=32 time<1ms TTL=128
Reply from 192.168.3.131: bytes=32 time<1ms TTL=128
Reply from 192.168.3.131: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



4- Conclusions:

The networking project successfully connects all 8 buildings of Al-Shorouk Academy with a scalable design, allocating 128 IPs per building to support current and future needs. This structured approach enhances connectivity, simplifies management, and strengthens security, providing a robust foundation for seamless operations, smart campus technologies, and improved academic and administrative efficiency.